Lab-10 Ex3 - Complex Number (PreLab*)

> *This is a PreLab exercise. Please do the PreLabs after each lecture and submit before you come to lab!*

You have learned struct in the lecture. It is useful in grouping related data together. Let's apply it to a very simple case – Complex Number.

Write a C program to read from the user *real* and *imaginary* (*i*) parts of two complex numbers, **`(a + bi)`** and **`(c + di)`**, and to store them using a structure defined in your code that stores the real and imaginary parts (both integer type).

Then the program will compute and output the results of addition and multiplication between the two complex numbers:

`(a + bi) + (c + di) = (a+c) + (b+d)i`

and

`(a + bi) * (c + di) = (ac-bd) + (bc+ad)i`

where `a`, `b`, `c` and `d` are **integers**.

The data structure `complex` and the function `printComplex()` are already provided for you in the code template in `main.c`.

```
struct complex {
    int real;
    int imgr;
} ;

void printComplex(struct complex x) {
     ...
}
```

Check out the structure and function to observe how the structure can be used, then use them to calculate the complex subtraction and multiplication!

Sample Run #1

2 2

1 1

`3 + 3i`

`4i`

Sample Run #2

9 7

7 1

16 + 8i

56 + 58i